

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

TASK NO.: 2

HTML5 Graphics and CSS3 Web Styling

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1. TITLE

HTML5 Graphics and CSS3 Web Styling

2. AIM

To understand how CSS3 enhances the appearance of web pages and to explore HTML5 graphics using the Canvas element.

3. OBJECTIVE

- To understand the use of HTML5 Canvas for drawing graphics.
- To understand CSS syntax and CSS selectors.
- To apply colors, fonts, borders, margins and padding.
- To use backgrounds, gradients, box shadows and text shadows.
- To create rounded corners using border-radius.
- To implement hover effects for buttons and navigation links.
- To design a webpage using a header, navigation bar, content cards and footer.

4. THEORY

CSS3 (Cascading Style Sheets) is used to control the appearance and layout of HTML elements. It allows a webpage to be styled using properties such as colors, fonts, backgrounds, borders, spacing, shadows, gradients and hover effects.

HTML5 Canvas provides a drawing area that can be controlled using JavaScript. In this task, the Canvas API is used to draw a rectangle, circle and line, and to display the name "Venu".

- HTML5 Canvas
- CSS syntax and selectors
- Colors and font families/sizes
- Margins and padding
- Borders and rounded corners
- Background colors and gradients
- Box shadow and text shadow
- Hover effects
- Flexbox layout for content cards

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document with the required head and body sections.
2. Add the page title and CSS rules inside the style element.
3. Create a header containing the webpage title and description.
4. Create a navigation bar with links to the webpage sections.
5. Create an introduction section with a heading, paragraph and button.
6. Add a Canvas element with a fixed width and height.

7. Use JavaScript Canvas methods to draw a rectangle, circle and line and display the name "Venu".
8. Create three content cards for HTML5, CSS3 and JavaScript.
9. Apply CSS properties including colors, fonts, borders, margins, padding, shadows and gradients.
10. Add hover effects to buttons and navigation links.
11. Create a footer containing the copyright and task information.
12. Open the HTML file in a web browser and verify the output.

File: task2.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>

  <style>
    /* Page styling */
    body {
      margin: 0;padding: 0;
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
    }
    /* Header */
    header {
      background: linear-gradient(to right, #355c7d, #6c9bcf);
      color: rgb(166, 159, 159);
      text-align: center;
      padding: 20px;
      text-shadow: 2px 2px 3px #b12929;
    }
    header h1 {
      font-size: 32px;
      font-family: "Trebuchet MS", sans-serif;
    }
    /* Navigation */
    nav {
      background-color: #333;
      padding: 12px;
      text-align: center;
    }
    nav a {
      color: white;
      text-decoration: none;
      margin: 0 15px;padding: 7px;
      font-size: 16px;
    }
    nav a:hover {
      color: #6c9bcf;
      background-color: white;
    }
  </style>
</head>
<body>
  <div>
    <div>
      <h1>My Webpage</h1>
    </div>
    <div>
      <a href="#home">Home</a>
      <a href="#about">About</a>
      <a href="#services">Services</a>
      <a href="#contact">Contact</a>
    </div>
  </div>
</body>
</html>
```

```

        border-radius: 5px;
    }
    /* Introduction */
    .intro {
        text-align: center;
        background-color: white;
        margin: 20px;padding: 20px;
        border: 5px solid #1d2ee1;
        border-radius: 50px;
        box-shadow: 3px 3px 8px #ea0d0d;
    }
    /* Buttons */
    button {
        margin-top: 10px;padding: 9px 16px;
        border: 1px solid #355c7d;
        border-radius: 6px;
        background-color: #355c7d;
        color: white;
        cursor: pointer;
    }
    button:hover {
        background-color: #1cc63b;
        transform: scale(1.20);
    }
    /* Canvas section */
    .canvas-section {
        text-align: center;
        background-color: #e6f2ff;
        margin: 20px;
        padding: 20px;
        border: 2px solid #9cc5e5;
        border-radius: 10px;
    }
    canvas {
        background-color: white;
        border: 2px solid #355c7d;
        outline: 2px dashed #d31895;
        border-radius: 6px;
        box-shadow: 10px 3px 8px #aaa;
    }
    /* Cards */
    .card-container {
        display: flex;
        justify-content: center;
        gap: 20px;
        padding: 30px;
        flex-wrap: wrap;
    }
    .card {
        background: linear-gradient(to bottom, white, #b0ba21);
        border: 2px solid #ccc;
        border-radius: 10px;
        padding: 20px;
        margin: 10px;
        width: 220px;
        box-shadow: 3px 3px 8px #aaa;
        text-align: center;
    }
    .card h3 {
        font-family: Georgia, serif;
        font-size: 23px;
        color: #355c7d;
    }
    /* Footer */
    footer {

```

```

        background-color: #333;
        color: white;
        text-align: center;
        padding: 15px;
        margin-top: 20px;
    }
</style>
</head>

<body>
    <!-- Main heading -->
    <header>
        <h1>My Styled Webpage</h1>
        <p>HTML5 Graphics and CSS3 Web Styling</p>
    </header>

    <!-- Navigation bar -->
    <nav>
        <a href="#home">Home</a>
        <a href="#canvas">Canvas</a>
        <a href="#cards">Technologies</a>
        <a href="#contact">Contact</a>
    </nav>

    <!-- Introduction -->
    <section id="home" class="intro">
        <h2>Welcome</h2>
        <p>This webpage demonstrates HTML5 Canvas and CSS3 styling.</p>
        <button>Explore More</button>
    </section>

    <!-- Part A: HTML5 Canvas -->
    <section id="canvas" class="canvas-section">
        <h2>Part A - HTML5 Canvas</h2>
        <p>The canvas contains a rectangle, circle, line and my name.</p>

        <canvas id="myCanvas" width="500" height="280"></canvas>

        <script>
            // Select canvas
            const canvas = document.getElementById("myCanvas");
            const ctx = canvas.getContext("2d");

            // Draw rectangle
            ctx.fillStyle = "skyblue";
            ctx.fillRect(40, 40, 130, 80);

            // Draw circle
            ctx.beginPath();
            ctx.arc(350, 80, 45, 0, 2 * Math.PI);
            ctx.fillStyle = "lightgreen";
            ctx.fill();
            ctx.stroke();

            // Draw line
            ctx.beginPath();
            ctx.moveTo(50, 170);
            ctx.lineTo(450, 170);
            ctx.strokeStyle = "skyblue";
            ctx.lineWidth = 3;
            ctx.stroke();

            // Display name
            ctx.font = "26px Arial";
            ctx.fillStyle = "darkblue";

```

```

        ctx.fillText("Venu", 220, 230);
    </script>
</section>

<!-- Part B: CSS3 Styling -->
<section id="cards">
    <h2 style="text-align:center;">Part B - CSS3 Styling</h2>

    <div class="card-container">

        <div class="card">
            <h3>HTML5</h3>
            <p>HTML5 provides the structure and content of a webpage.</p>
            <button>Learn HTML</button>
        </div>

        <div class="card">
            <h3>CSS3</h3>
            <p>CSS3 is used to style webpages with colors, fonts and layouts.</p>
            <button>Learn CSS</button>
        </div>

        <div class="card">
            <h3>JavaScript</h3>
            <p>JavaScript adds interaction and functionality to webpages.</p>
            <button>Learn JS</button>
        </div>

    </div>
</section>

<!-- Footer -->
<footer id="contact">
    <p>&copy; 2026 My Styled Webpage</p>
    <p>HTML5 & CSS3 Observation Task</p>
</footer>

</body>

</html>

```

7. CODE EXPLANATION

Element / Property	Purpose
body	Sets the overall page margin, padding, font and background color.
header	Creates the main header with gradient background and text shadow.
nav and nav a	Styles the navigation bar and links.
nav a:hover	Changes navigation link appearance on hover.
.intro	Uses background, border, rounded corners and box shadow for the introduction.
button and button:hover	Styles buttons and changes their appearance and scale on hover.
<canvas>	Provides the HTML5 drawing area.
getContext("2d")	Obtains the 2D drawing context.
fillRect()	Draws the rectangle.
arc()	Creates the circle path.
moveTo() and lineTo()	Define the line.
fillText()	Displays the name "Venu" on the Canvas.

.card-container	Uses Flexbox to arrange the cards.
.card	Styles each technology card using gradient, border, padding, margin and shadow.
footer	Styles the bottom section of the webpage.

8. TEST CASES AND EXECUTION DATA

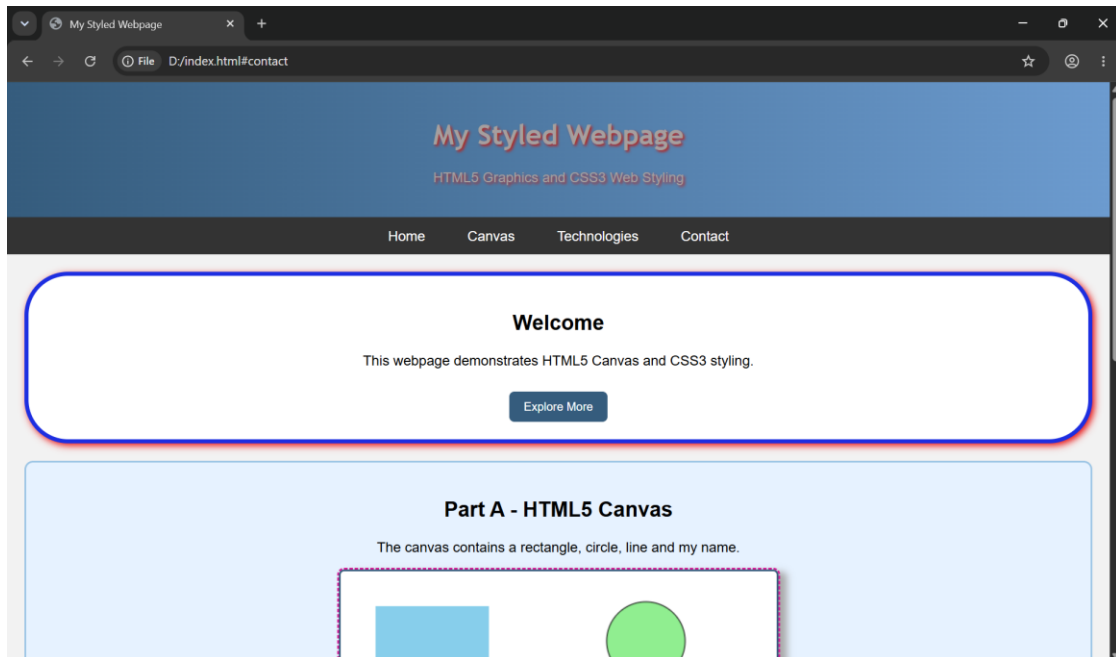
The following test cases can be used to verify the major requirements. Fill the actual result/status after checking the webpage in a browser.

Test Case	Test / Input	Expected Result	Actual Result / Status
TC-01	Open the HTML file	Header, navigation and introduction should be displayed	To be filled
TC-02	Check Canvas	Rectangle, circle, line and "Venu" should appear	To be filled
TC-03	Check CSS styling	Colors, fonts, borders, rounded corners, spacing and shadows should be visible	To be filled
TC-04	Check cards	Three cards for HTML5, CSS3 and JavaScript should be displayed	To be filled
TC-05	Check button hover	Button should change appearance and scale on hover	To be filled
TC-06	Check navigation hover	Navigation link should change color/background on hover	To be filled
TC-07	Check footer	Footer should be displayed with task information	To be filled

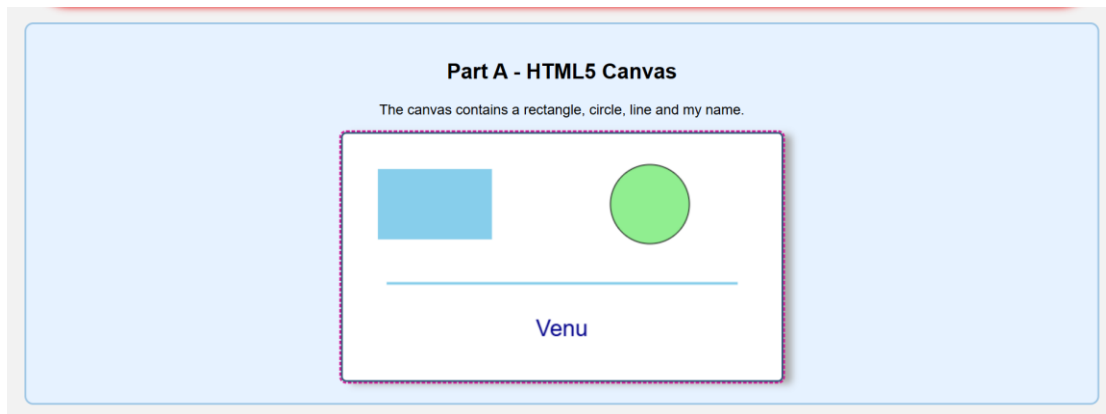
9. OUTPUT

Paste the actual browser screenshots in the spaces below.

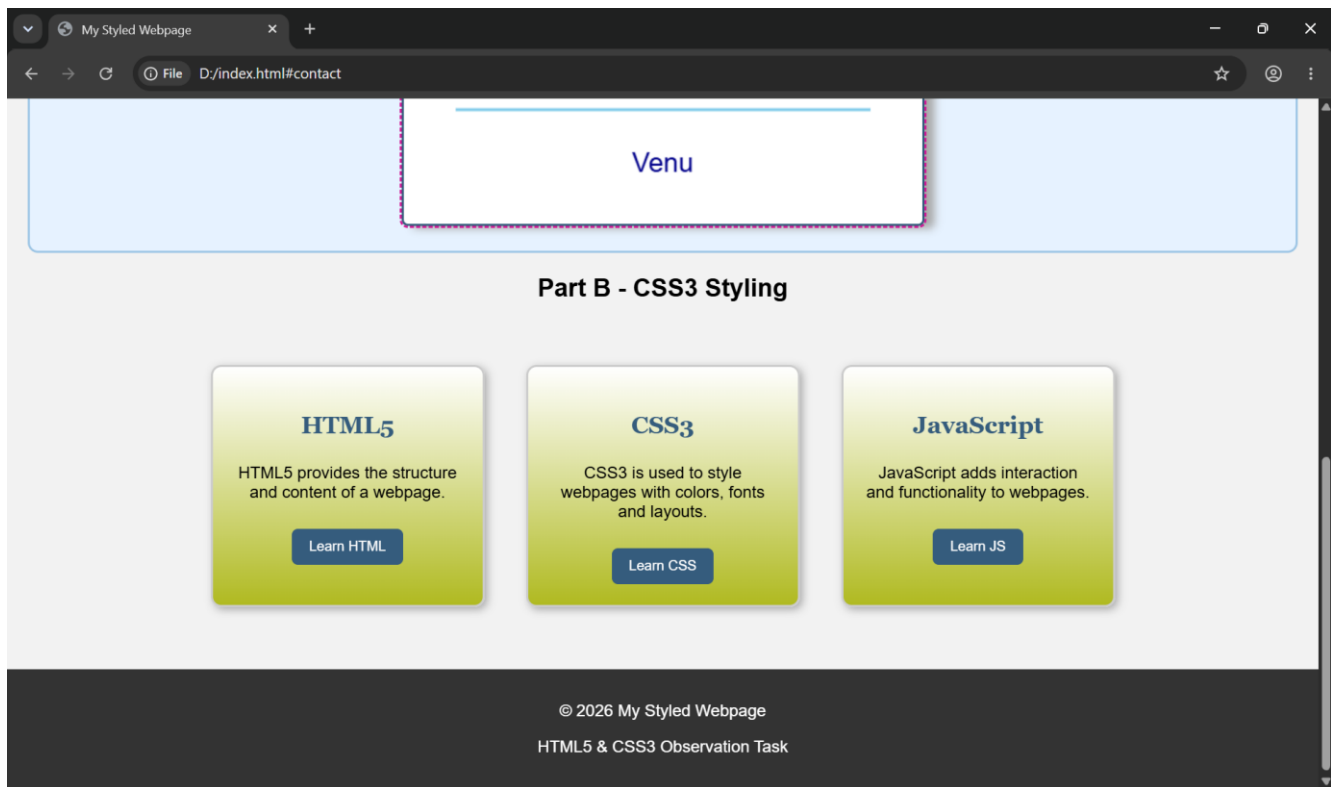
Output 1: Complete Styled Webpage



Output 2: HTML5 Canvas – Rectangle, Circle, Line and Name



Output 3: Three CSS3 Technology Cards



Output 4: Button / Navigation Hover Effect(After cursor is placed on button)



10. RESULT

Thus, a styled webpage titled "My Styled Webpage" was developed using HTML5 and CSS3. The Canvas element was used to draw a rectangle, circle and line and to display the name "Venu". CSS3 properties such as colors, fonts, borders, margins, padding, gradients, shadows, rounded corners and hover effects were applied to design the webpage with a header, navigation bar, three content cards and a footer.

11. SUBMISSION CONTENT

- HTML source file containing the complete webpage, CSS and Canvas JavaScript.
- Browser output screenshots inserted into the observation record.
- Final observation record converted to PDF, if required by the faculty.
- Source code and record maintained in the GitHub repository.